

REDA WAFIK

Data Engineering & AI | Data Pipelines | Machine Learning
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Profile

Data Engineering and AI graduate with hands-on experience designing and building data-driven systems across data warehousing, ETL orchestration, real-time streaming, and distributed systems, complemented by machine learning, natural language processing, and computer vision project work. Experience includes dimensional modeling, pipeline orchestration with Airflow, message-driven architectures with Kafka and RabbitMQ, stream processing with Flink and Spark, and LLM-based natural-language querying. Interested in building scalable, reliable data infrastructure and real-time analytics systems.

Education

FST Tangier 2026
Master in Artificial Intelligence and Data Science (AIDS)

FST Settat 2024
Bachelor's Degree in Computer Engineering

Professional Experience

Final-Year Project Intern, TE Connectivity Feb 2026 – Jul 2026
Integrated Business Intelligence Platform for Manufacturing and Finance KPIs

- Designed a dimensional PostgreSQL Data Warehouse using a star schema with snapshot-based historization to model CIP and Fixed Assets data for structured decision-support analytics.
- Orchestrated automated ETL pipelines with Apache Airflow, distributing monthly SAP ERP data extraction, transformation, and KPI computation across RabbitMQ and Celery workers.
- Engineered a FastAPI backend with role-based access control and Redis-cached KPI retrieval, powering interactive BI dashboards for manufacturing and finance KPI monitoring.
- Developed an AI assistant integrating LangChain with a self-hosted Ollama LLM for natural-language Text-to-SQL querying, enforcing deterministic SQL validation for read-only, domain-restricted, and prompt-injection-resistant access.

Data Engineer Intern, TE Connectivity Jul 2025 – Sep 2025
Quality and Production Real-Time Monitoring Application

- Analyzed an existing HYDRA MES, AWS Lambda, and Aurora DB batch pipeline based on two-minute polling and designed a four-layer distributed architecture covering ingestion, streaming, storage, and visualization.
- Implemented a clustered RabbitMQ ingestion layer and a three-broker Apache Kafka streaming layer publishing machine data, quality metrics, and production KPIs into partitioned topics.
- Deployed an Apache Flink cluster with JobManagers and three TaskManagers to compute real-time KPI aggregations including cycle time, temperature, and scrap rate.
- Implemented a hybrid storage architecture using Redis for recent KPI data and Cassandra for historical data, and delivered real-time Grafana dashboards with threshold-based anomaly alerts.
- Automated deployment and orchestration using Python and Docker Compose.

Full-Stack Developer Intern Apr 2025 – Jun 2025
E-Commerce Platform for a Bookstore – Admin Panel and Client Section

- Designed and developed a two-sided e-commerce platform with Laravel, React, and Tailwind CSS to display and sell bookstore products.
- Built an Admin Panel covering order management, product and category management, sales reporting, and automatic invoice generation.
- Built a Client Section featuring category-based product browsing, product detail pages, a shopping cart, order placement with tracking, automatic invoice access, and user account management.
- Delivered a fully responsive interface across desktop, tablet, and mobile devices with a focus on smooth navigation and usability.

Academic Projects

Real-Time Amazon Customer Review Sentiment Analysis
Apache Kafka, Spark Streaming, PySpark MLlib, NLP

- Built a real-time NLP pipeline ingesting Amazon customer reviews through Apache Kafka and performing preprocessing, lemmatization, TF-IDF feature engineering, and train-validation-test splitting.
- Trained and compared supervised classification models using PySpark MLlib and selected the best-performing model based on validation performance.
- Developed a real-time prediction interface with an offline MongoDB-backed dashboard for monitoring sentiment pre-

dictions.

Content-Based Image Retrieval with YOLO

Computer Vision, Object Detection, Feature Extraction

- Designed a content-based image retrieval system using YOLOv8n for object detection and localization across a 15-category image collection.
- Developed a feature extraction pipeline combining color histograms, Tamura and Gabor texture descriptors, and Hu moments.
- Exposed the retrieval system through a Flask and Flask-RESTful API consumed by a MEAN-stack web application.
- Implemented similarity-search and ranking workflows for retrieving visually similar detected objects.

AI Linux CMD Assistant

Fine-Tuned LLM, PEFT (LoRA), FastAPI, React

- Built an AI assistant providing help and explanations for Linux commands, powered by a Llama 2 model fine-tuned with PEFT (LoRA) for specialized Linux knowledge.
- Developed a FastAPI backend serving the fine-tuned model and handling API requests, with automatic GPU detection and CPU fallback for inference.
- Built a user-friendly web interface with React and Vite, styled with Tailwind CSS, for interacting with the assistant.
- Implemented the fine-tuning pipeline in a Jupyter Notebook using Hugging Face Transformers, PEFT, PyTorch, and Pandas.

Technical Skills

Data Engineering and Distributed Systems: Apache Kafka, Apache Spark, Apache Flink, Apache Airflow, RabbitMQ, Celery, ETL, Real-Time Streaming, Distributed Systems, n8n

Artificial Intelligence and Machine Learning: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Supervised Learning, Object Detection, Feature Extraction, Scikit-learn, TensorFlow, PyTorch, PySpark MLlib, LangChain, Ollama, Text-to-SQL

Programming and Query Languages: Python, SQL, JavaScript

Databases and Data Stores: PostgreSQL, MongoDB, Cassandra, Redis

Web, APIs and Infrastructure: FastAPI, Flask, Angular, Docker, Docker Compose, REST APIs

Certifications

- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI, Feb 2025
- **Fundamentals of Deep Learning** – NVIDIA, Nov 2025
- **SAP FICO (Finance and Controlling)** – Udemy Course Completion, Aug 2026

Languages

Arabic – Native | French – Professional | English – Professional